

Diaphragm Seal (DS)

Diaphragm seals are used when the media can falsify the pressure measurement due to high viscosity, corrosive media or their propensity to crystalize. These seals transmit the process pressure to the measuring instrument with the diaphragm, forming a hermetic seal between the media and the measuring instrument. Available with various process connection, diaphragm seals are especially suitable for general process and engineering applications.

The diaphragm seals are designed to match the dimensions of standardised flanges (ANSI B 16.5, DIN, etc). The diaphragm is either internal or open to process, depending on the design.

The wetted parts of these seals are manufactured in stainless steel as the standard version. Bourdon tube pressure gauge or transmitters, are suitable for pressure ranges upto 400 kg/sq cm.

For corrosive applications, the wetted parts in contact with the media can be manufactured with special materials such as Monel, Hastalloy and Tantalum.

Available In

Pressure range	0 to 400 Kg/Sq.cm
Vacuum range	1 to 0 Kg/Sq.cm
Compound range	1 to 39 Kg/Sq.cm

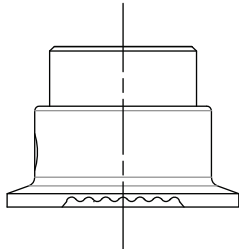
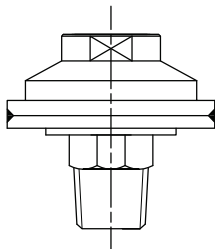
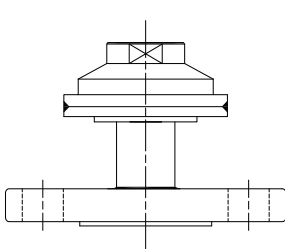
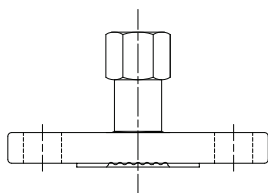
Special Features

- Variety of process connections
- Overload protection by diaphragm bed
- Change of process connection does not disturb the measurement characteristics
- Special materials for corrosive applications
- Mountable on instruments and control equipment
- Can be offered with optional capillary extension



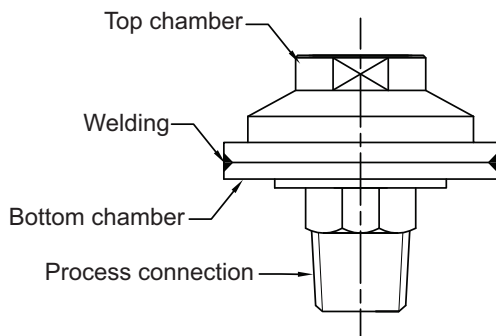
Industrial Applications

- Chemical process industries
- Thermal, nuclear and hydroelectric power stations
- Food process industries
- Iron and steel industries
- Rubber moulding / processing plants
- Sugar industries
- Paper industries
- Petroleum industries
- General mechanical engineering industries

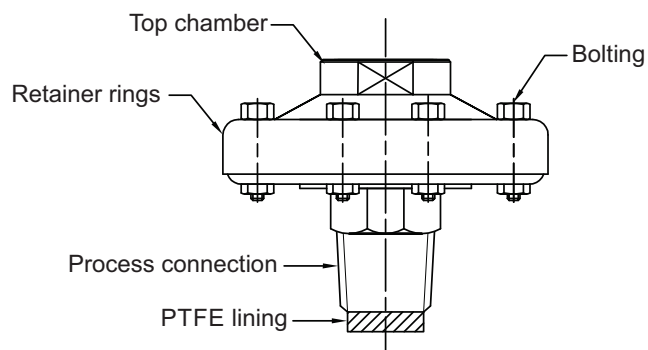
Standard Product Specifications				
MODEL	Triclover	Screw-End	Flanged-End	Flush type-Flanged End
Code	T	S	F	R
Images				
Instrument Connection	3/8" BSP (standard) / 1/2" NPT (optional)			
Process Connection Size	1" 1.5" 2" 2.5"	M 20 X 1.5 P M 27 X 2 P M 33 X 2 P 1/4" 3/8" 1/2" 3/4" 1" Pipe welding (threads Not required)	1/2" 3/4"	1" 1.5" 2" 2.5" 3"
Process Connection Type	Triclover seal Triclover seal with clamp	BSP NPT BSPT	#150 RF/FF #300 RF/FF #600 RF #900 RF/RTJ #1500 RTJ #2500 RTJ	
Top Chamber MOC	SS 304 (standard) SS 316L			
Diaphragm MOC	SS 316L	SS 316L (standard) Hastelloy C276 Monel 400 Inconel 825 Tantalum Titanium Nickel		
Bottom Chamber/ Flange MOC	Nil	SS 316L (standard) Hastelloy C276 Monel 400 Inconel 825	SS 304 (Standard) SS 316 SS 316L	
PTFE Protection (optional)	Nil	PTFE sheet for diaphragm PTFE lining for bottom chamber/flange PTFE sheet for diaphragm and lining for bottom chamber/ flange PTFE coating for diaphragm and bottom chamber/flange		
Filling Fluid	Silicon oil - process temperature -10 to 120 Deg C (standard) DC 704 - process temperature 0 to 220 Deg C Food grade oil - process temperature -25 to 75 Deg C (standard for triclover)			
Capillary Length (optional)	Nil	Min - 1 meter to max - 5 meters (SS 316L micro-bore capillary with SS 304 armour as standard)		
Manufacturing Standard	EN 837 - 1			

Mounting Options

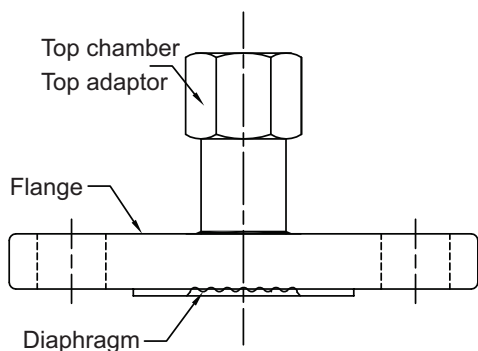
Seal with Thread End



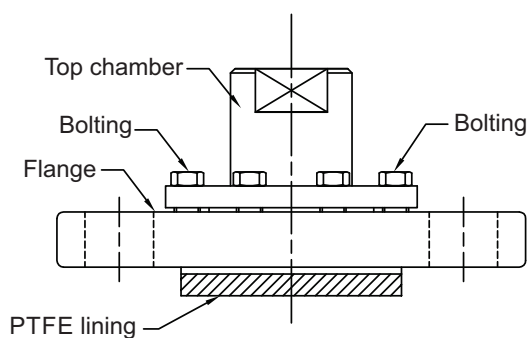
Seal with Thread End with PTFE Lining (Working Pressure ≤ 40 bar)



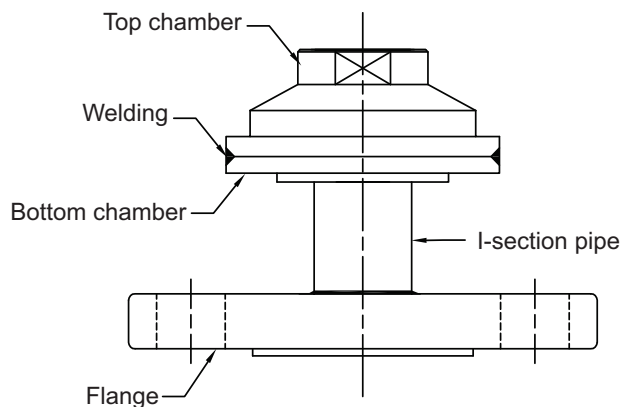
Seal with Flange End - Flush Type (For Flange connections ≥ 1 " size)



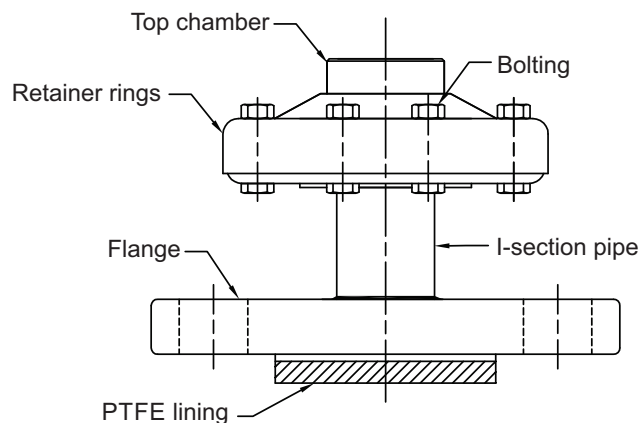
Seal with Flange End - Flush Type with PTFE Lining (For Flange connections ≥ 1 " Size & Working Pressure ≤ 40 bar)



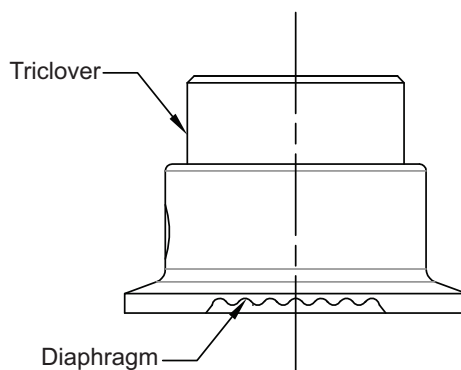
Seal with Flange End - I-Section Type (For Flange connections < 1 " size)



Seal with Flange End - I-Section Type with PTFE Lining (For Flange connections < 1 " Size & Working Pressure ≤ 40 bar)



Seal with Triclover End



Configuration Generation Code

Top Chamber
SS 304 (standard)
SS 316

Diaphragm - Material of Construction			
Screwed End	Flanged End	Flush Type Flanged End	Triclover End
SS 316L (standard)	SS 316L (standard)	SS 316L (standard)	SS 316L
Monel 400	Monel 400	Monel 400	-NA-
Hastelloy C276	Hastelloy C276	Hastelloy C276	
Inconel 825	Inconel 825	Inconel 825	
Tantalum	Tantalum	Tantalum	
Titanium	Titanium	Titanium	
Nickel	Nickel	Nickel	

Bottom Chamber / Flange - Material of Construction			
Screwed End	Flanged End	Flush Type Flanged End	Triclover End
SS 316L (standard)	SS 304 (standard)	SS 304 (standard)	-NA-
Hastelloy C276	SS 316	SS 316	
Monel 400	SS 316L	SS 316L	
Inconel 825			

Optional Features
Flange type as per ASME B16.5
Retainer rings
Flushing rings
Special requirement – please specify

Ordering Information

**

*

*

*

*

**

*

*

Diaphragm Seal Type	Code
Screwed end	S
Flanged end	F
Flanged end flush type	R
Triclover end	T

Instrument Connection Size	Code
3/8" (standard)	B
1/2"	C

Instrument Connection Type	Code
BSP(F) (standard)	E
NPT(F)	D

Process Connection Size							
Screwed End	Code	Flanged End	Code	Flanged End Flush Type	Code	Triclover End	Code
M 20X1.5 P	1	1/2"	C	1"	E	1"	E
M 27X2 P	2	3/4"	D	1.5"	G	1.5"	G
M 33X2 P	3			2"	H	2"	H
1/4"	A			2.5"	I	2.5"	I
3/8"	B			3"	J		
1/2"	C						
3/4"	D						
1"	E						
Pipe welding (threads not required)	Z						

Process Connection Type							
Screwed End	Code	Flanged End	Code	Flanged End Flush Type	Code	Triclover End	Code
BSP(M)	A	#150 RF	1	#150 RF	1	Triclover seal	W
NPT(M)	B	#150 FF	7	#150 FF	7	Triclover seal with clamp	X
BSPT(M)	C	#300 RF	2	#300 RF	2		
		#300 FF	8	#300 FF	8		
		#600 RF	3	#600 RF	3		
		#900 RF	4	#900 RF	4		
		#900 RTJ	H	#900 RTJ	H		
		#1500 RTJ	I	#1500 RTJ	I		
		#2500 RTJ	J	#2500 RTJ	J		

Fluid Filling	
Fluid	Code
Silicon oil - process temperature -10 to 120 Deg C (standard)	1
DC 704 - process temperature 0 to 220 Deg C	2
Food grade oil - process temperature -25 to 75 Deg C (standard for Triclover)	3

Diaphragm and PTFE Protection	Code
Without PTFE protection	1
PTFE sheet for diaphragm	2
PTFE lining for bottom chamber / flange	3
PTFE sheet for diaphragm and lining for bottom chamber / flange	4
PTFE coating for diaphragm and bottom chamber / flange	5

Capillary Length and Material of Construction	Code
Not required	00
1M. SS 316L micro-bore capillary with SS 304 armour (standard)	01
1M. SS 316L micro-bore capillary with SS 304 armour and PVC	02
2M. SS 316L micro-bore capillary with SS 304 armour	03
2M. SS 316L micro-bore capillary with SS 304 armour and PVC	04
3M. SS 316L micro-bore capillary with SS 304 armour	05
3M. SS 316L micro-bore capillary with SS 304 armour and PVC	06
4M. SS 316L micro-bore capillary with SS 304 armour	07
4M. SS 316L micro-bore capillary with SS 304 armour and PVC	08
5M. SS 316L micro-bore capillary with SS 304 armour	09
5M. SS 316L micro-bore capillary with SS 304 armour and PVC	10
General note : PTFE protection and capillary is not applicable for Triclover model	

Ordering example: SBECB1100-(0000) (with standard selection)



Forbes Marshall
Krohne Marshall
Forbes Marshall Arca
Codel International
Forbes Solar
Forbes Vyncke
Forbes Marshall Steam Systems

Plot No. A-19/2 & T-4/2, I.D.A
Nacharam, Hyderabad - 500 076
Tel : +91-40-27179223, 27171909,
27170732
Fax : +91-40-27173235

Email : sales-gauges@forbesmarshall.com, ccmidc@forbesmarshall.com

© All rights reserved. Any reproduction or distribution in part or as a whole without written permission of Forbes Marshall Pvt Ltd, its associate companies or its subsidiaries ("FM Group") is prohibited.
Information, designs or specifications in this document are subject to change without notice. Responsibility for suitability, selection, installation, use, operation or maintenance of the product(s) rests solely with the purchaser and/or user. The contents of this document are presented for informational purposes only. FM Group disclaims liabilities or losses that may be incurred as a consequence of the use of this information.

B-85, Phase II, Chakan Indl Area
Sawardari, Chakan, Tal. Khed
Dist. Pune - 410 501. INDIA
Tel : 91(0)2135-393400

A-34/35, MIDC H Block
Pimpri, Pune - 411 018. INDIA.
Tel : 91(0)20-27442020, 39851199
Fax : 91(0)20-27442040

CIN No.: U28996PN1985PTC037806
www.forbesmarshall.com